FINAL YEAR PROJECT PROPOSAL

AR-BASED INTERACTIVE SURGICAL TRAINING SYSTEM WITH REAL-TIME HAPTIC EMULATION

INU

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A project proposal submitted to The IQRA National University, Peshawar in partial fulfillment of the requirements for the degree of

BACHELOR OF SCIENCE (BS) IN COMPUTER SCIENCE

SCHOOL OF COMPUTER SCIENCE

IQRA NATIONAL UNIVERSITY, PESHAWAR-PAKISTAN

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List of Acronyms

Acronym	Description
INU	IQRA National University
AI	Artificial Intelligence
NLP	Natural Language Processing
LLM	Large Language Model
API	Application Programming Interface
UI / UX	User Interface / User Experience
BLE / IoT	Bluetooth Low Energy / Internet of Things

1. Introduction & Problem Statement

The project titled **"AR-Based Interactive Surgical Training System with Real-Time Haptic Emulation"** addresses critical architectural paradigms within the modern computer science landscape. As information engineering systems shift toward distributed, agile environments, specialized automated monitoring and optimization models become absolutely paramount to unlock next-generation performance.

Problem Statement: Medical students encounter extreme limitations when trying to obtain physical access to cadavers or real operating room practice windows. Traditional text summaries and video modules fail to cultivate muscle memory or convey structural physical spatial feedback necessary for managing high-risk surgical errors. Without an automated, reliable system structure to streamline these

processes, operational risks escalate dramatically. This system solves this missing gap by developing an optimized computational pipeline that acts as a robust mechanism for resolving these complex challenges.

2. Objectives of Project

The primary structural milestones mapped for this software development cycle are:

- Construct highly accurate, interactive 3D anatomical organ models exhibiting dynamic real-time physical deformation responses.
- Implement an Augmented Reality overlay system mapping virtual spatial assets onto target boundaries accurately via consumer mobile cameras.
- Integrate BLE haptic actuator feedback control systems communicating precise resistance metrics directly to user devices.

3. Methodology & System Architecture

The system adheres to a structured, modular pipeline layout designed using a decoupled Client-Server model. This ensures that heavy backend model inferencing jobs do not impair the frontend interface application execution parameters.

Core Modules:

1. **Authentication & Profile Integration Module:** Handles user record enrollment and data secure caching sequences smoothly.
2. **Core Execution and Inference Pipeline:** Manages real-time data ingestion, transformation operations, and algorithmic processing nodes.
3. **Analytics & Reporting Layout Engine:** Aggregates calculation products, compiling structured system reports safely.

[System Architecture Design: React Native Client Interface ←→ Secure Gateway APIs ←→ Python/ Node Compute Processing Engines ←→ Scalable Vector/Relational Databases]

4. Proposed Time Schedule

Phase	Activity Description	Duration (2025-2026)
Phase 1	Literature Survey & Component Finalization	December
Phase 2	Architecture Modeling & Engineering Checks	January - February
Phase 3	Core Engine Development & Coding Cycle	March - April
Phase 4	System Verification Testing & Report Synthesis	May - June

Visual Timeline (Gantt Chart Representation)

Activity	Dec	Jan	Feb	Mar	Apr	May
Literature Gathering	■					
Scheme Design Modeling		■	■			
System Core Coding				■	■	
Verification & Writing						■

5. Budget Description & Technical Stack

The project leverages open-source libraries and public cloud deployment tiers, achieving maximum software efficiency with minimal capital expenditures. All computation and model runs are hosted on standard developer hardware platforms already in position.

Proposed Technical Core Stack: React Native Bridge, Unity 3D Engine, ARCore Framework, C# Scripting, BLE Serial Protocols, Arduino Micro-Firmware.

6. Undertaking & Project Declarations

We hereby declare that this technical development proposal is our genuine work executed under supervision at IQRA National University. No portion of this engineering research layout has been submitted previously for academic credit at any alternative educational board.

Student 1 Signature

Student 2 Signature

7. Institutional Review Panel Logs

Supervisor Evaluation Section:

_____

Supervisor Signature Date: ____/____/_____

FYP Department Coordinator Workspace:

_____

Coordinator Signature Status: APPROVED / HOLD